Rational Inattention and Endogenous Volatility: A Large Deviation Approach

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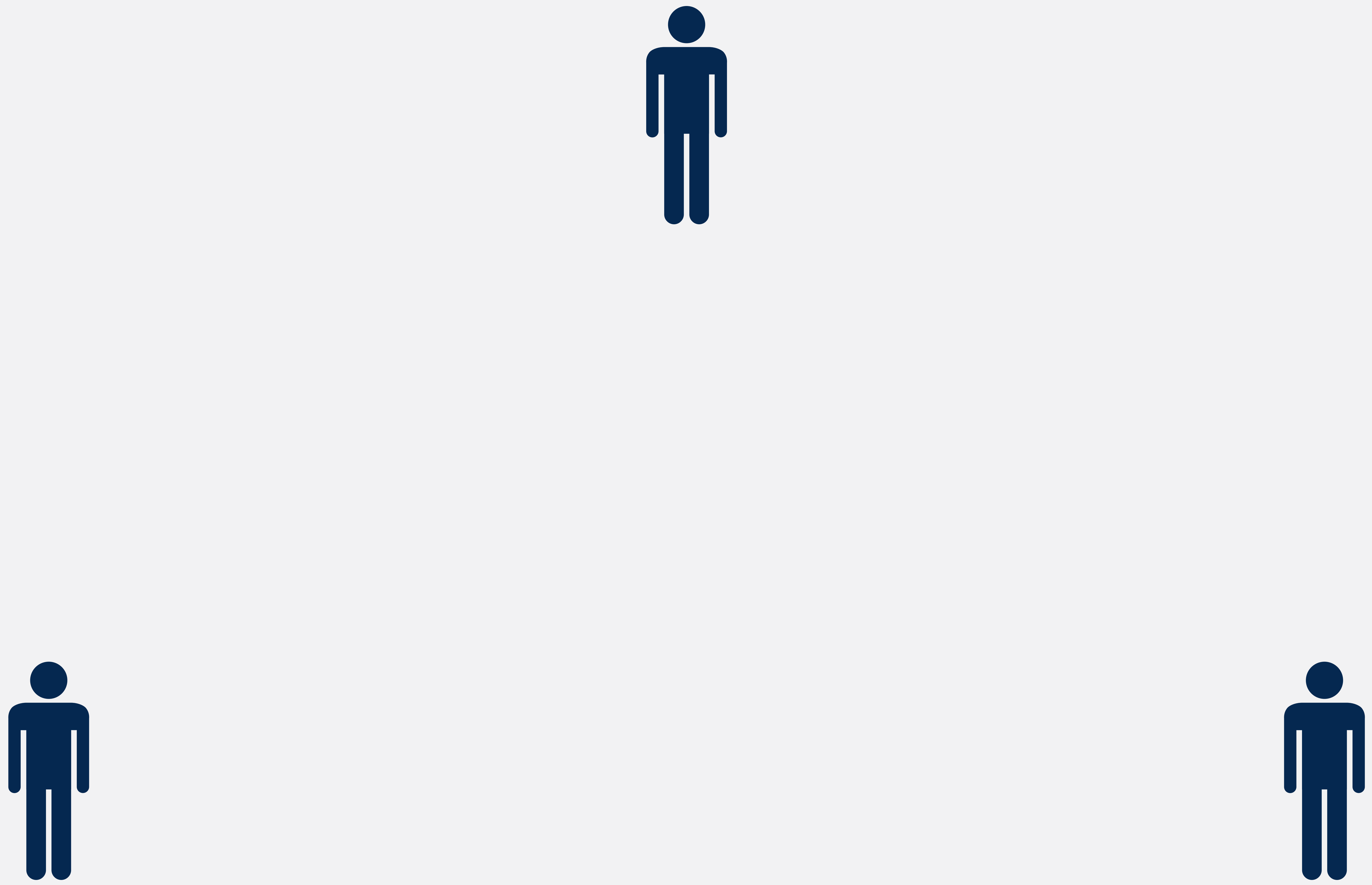
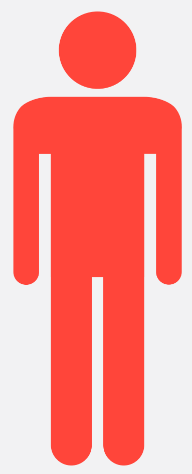


invest ($a_i = 1$) or not invest ($a_i = 0$)



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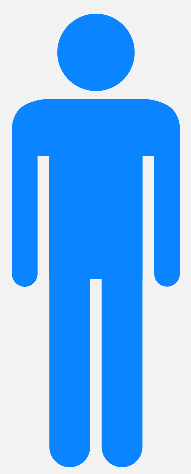
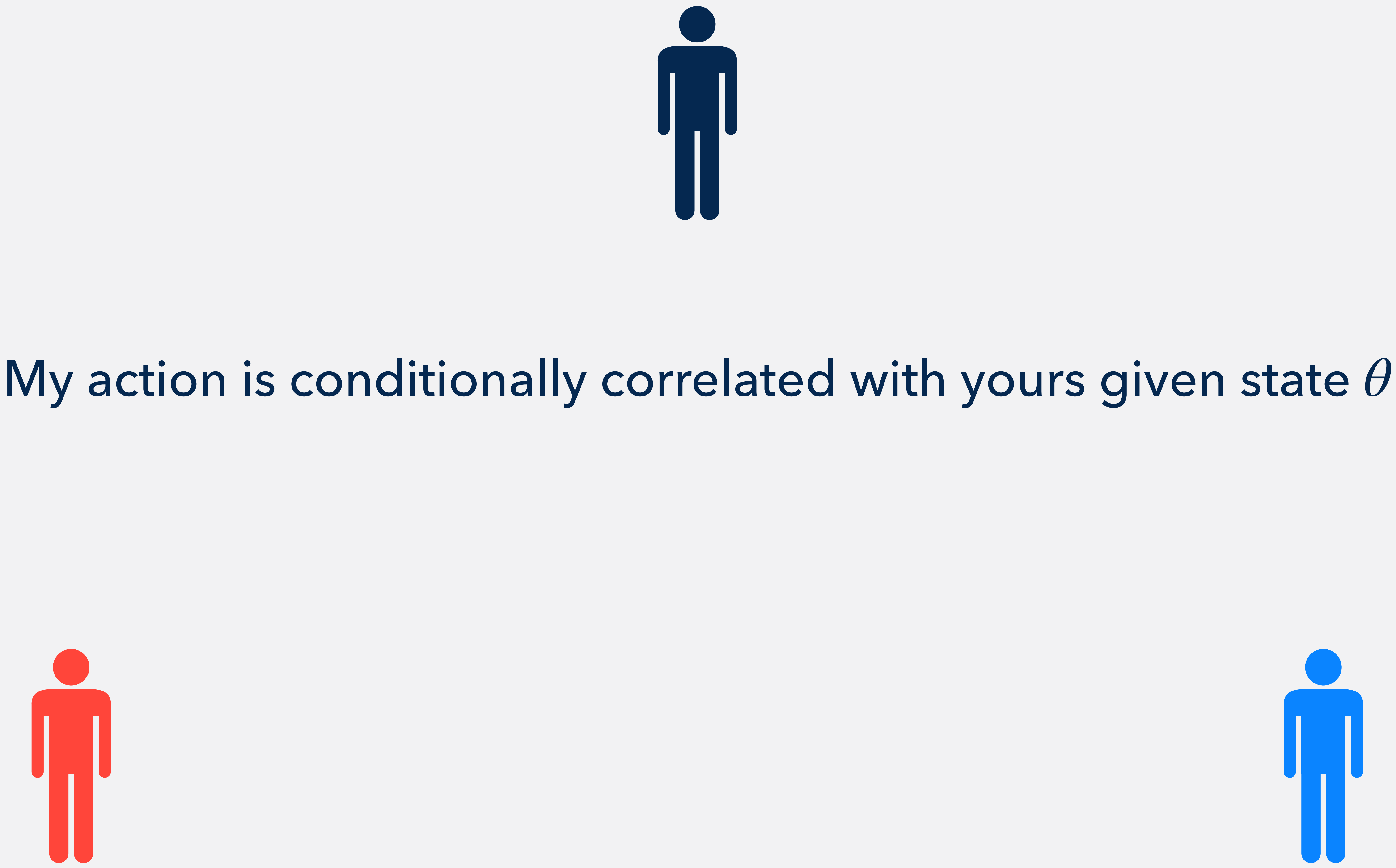


I am uncertain about

both **exogenous** θ and **endogenous** $\bar{a}_n$

I will pay attention to

both **exogenous** θ and **endogenous** $\bar{a}_n$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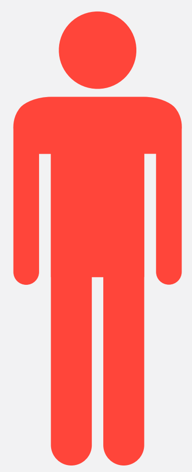

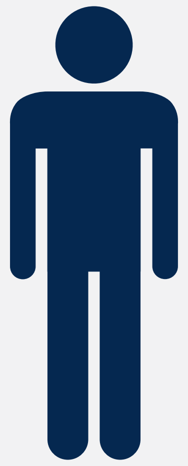

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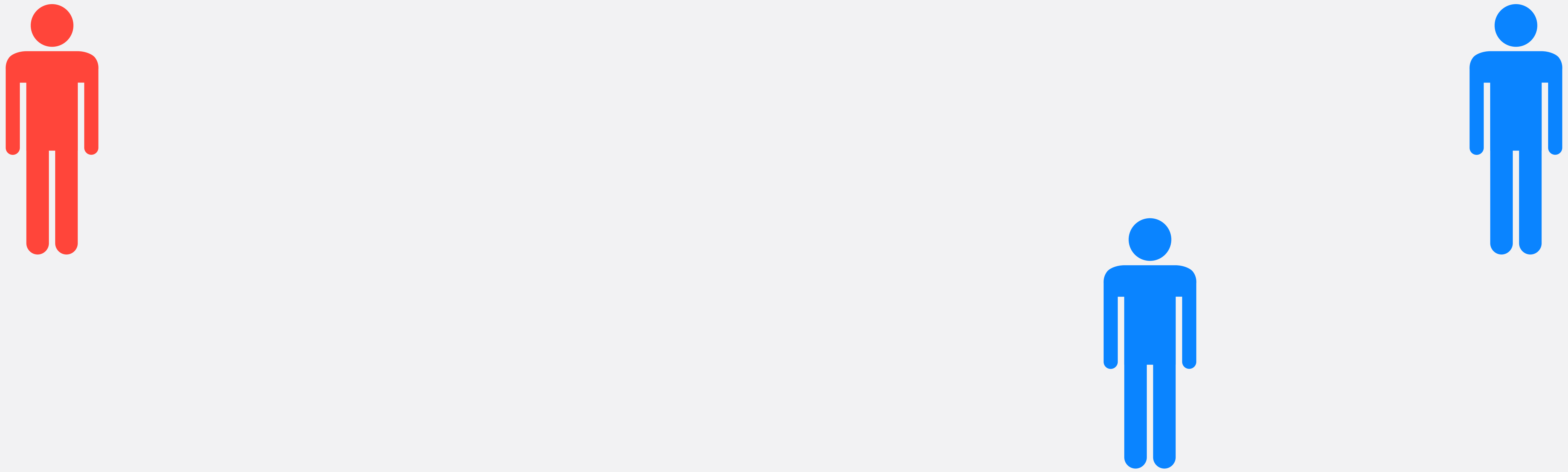

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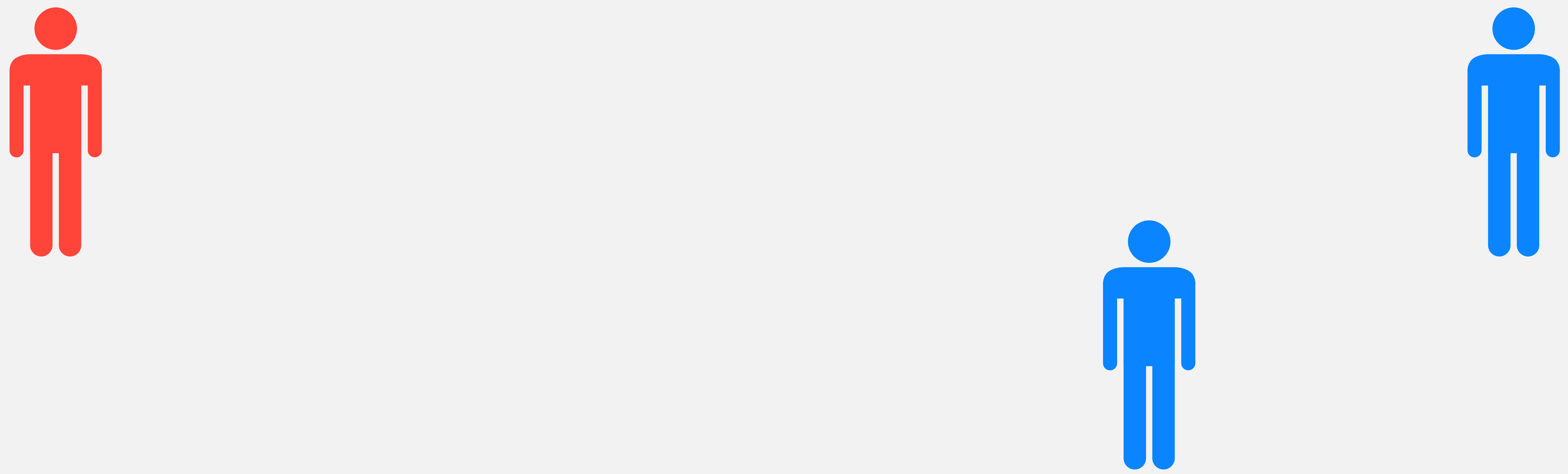

What if #agents → ∞?

How does **information acquisition** affect aggregate behavior in large games?

rational inattention

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Can such aggregate behavior be described by a **representative agent** model?

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endogenous volatility?

Endogenous Volatility

$$\mathrm{Var}\left(\bar{a}_n\right) = \mathrm{Var}\left(\mathbb{E}\left[\bar{a}_n \mid \theta\right]\right) + \mathbb{E}\left[\mathrm{Var}\left(\bar{a}_n \mid \theta\right)\right]$$

Endogenous Volatility

exogenous volatility

variations of the state

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endogenous volatility

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$$\mathrm{Var}\left(\bar{a}_n \mid \theta\right) = \frac{1}{n}\mathrm{Var}\left(a_i \mid \theta\right) + \frac{n-1}{n}\mathrm{Cov}\left(a_i, a_j \mid \theta\right)$$

Endogenous Volatility

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$$\mathrm{Var}\left(\bar{a}_n \mid \theta\right) \approx \qquad \mathrm{Cov}\left(a_i, a_j \mid \theta\right)$$

#agents n is large

Key modeling features

- aggregative potential games
  - beauty contests, bank runs, voting, congestion, etc.
- rational inattention with entropy cost

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Overview

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Not just the Law of Large Numbers!

LLN needs Cov$\left(a_i, a_j \mid \theta\right) \approx 0$, which *is* a result

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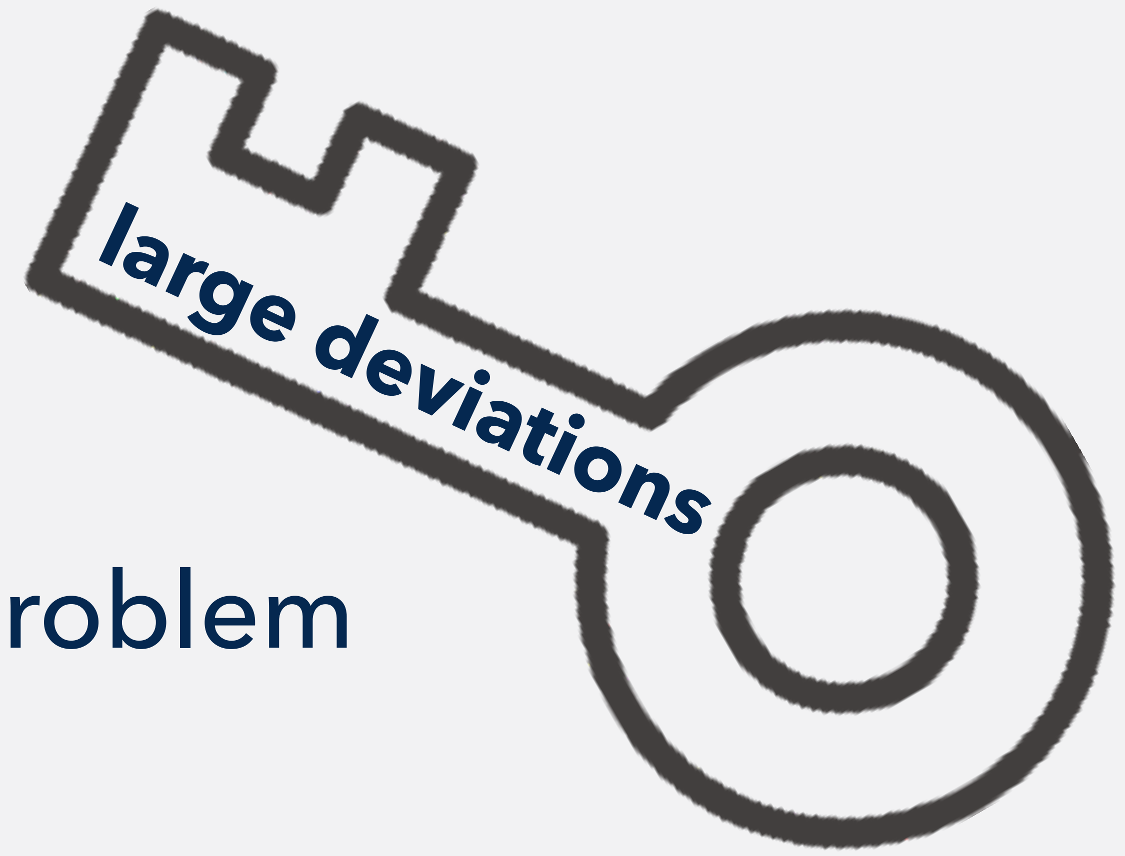

Example:

Key modeling features

Large-deviations logic behind main results

Main Results

General settings

Statements

Base Game

State $\theta \sim$ common prior $\pi \in \Delta(\text{finite } \Theta)$

Agent $i = 1,\ldots,n$ chooses action $a_i \in \{1,0\}$

$$u_i(a,\theta) = \begin{cases} \bar{a}_n - \theta & \text{if } a_i = 1 \\ 0 & \text{if } a_i = 0 \end{cases}$$

potential

$$v_n(a,\theta) = n\left[\frac{\bar{a}_n^2}{2} - \theta\,\bar{a}_n\right] + \frac{\bar{a}_n}{2}$$

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Definition: $u_i(a_i', a_{-i}, \theta) - u_i(a_i, a_{-i}, \theta)$

$= v_n(a_i', a_{-i}, \theta) - v_n(a_i, a_{-i}, \theta)$

thus, maximizing $u_i \Longleftrightarrow$ maximizing v_n

state

θ

actions

$a_1, \ldots, a_n$

average action

$\bar{a}_{-i} = \frac{1}{n-1} \sum_{j \neq i} a_j$

Agent i optimizes over conditional action distributions $P_i\left(a_i \mid \bar{a}_{-i}, \theta\right)$:

$$\max_{P_i} \ \mathbb{E}\left[u_i(a_i, a_{-i}, \theta)\right] - \text{information cost}$$

Rational Inattention

Sims 2003 Matějka–McKay 2015 Caplin–Dean–Leahy 2021 Denti 2023

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Decision-making process

- choose signal distributions $P_i(x_i \mid \bar{a}_{-i}, \theta)$
- observe a signal and update a belief
- choose an action a_i under the posterior

Assume wlog that she directly chooses $P_i(a_i \mid \theta)$

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Information cost

$$\text{entropy cost} \;=\; \text{unit cost } \lambda \;\times\; \text{uncertainty reduction in terms of entropy}$$

more

state
θ

$\xrightarrow{\text{information}}$

actions
$a_1, \ldots, a_n$

$\xleftarrow{\text{information}}$

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more

$$\boxed{\begin{array}{c}\textbf{state}\\ \theta\end{array}} \xrightarrow{\text{information}} \boxed{\begin{array}{c}\textbf{actions}\\ a_1, \dots, a_n\end{array}} \xleftarrow{\text{information}} \boxed{\begin{array}{c}\textbf{average action}\\ \bar{a}_{-i} = \frac{1}{n-1}\sum_{j\neq i} a_j\end{array}}$$

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more

state

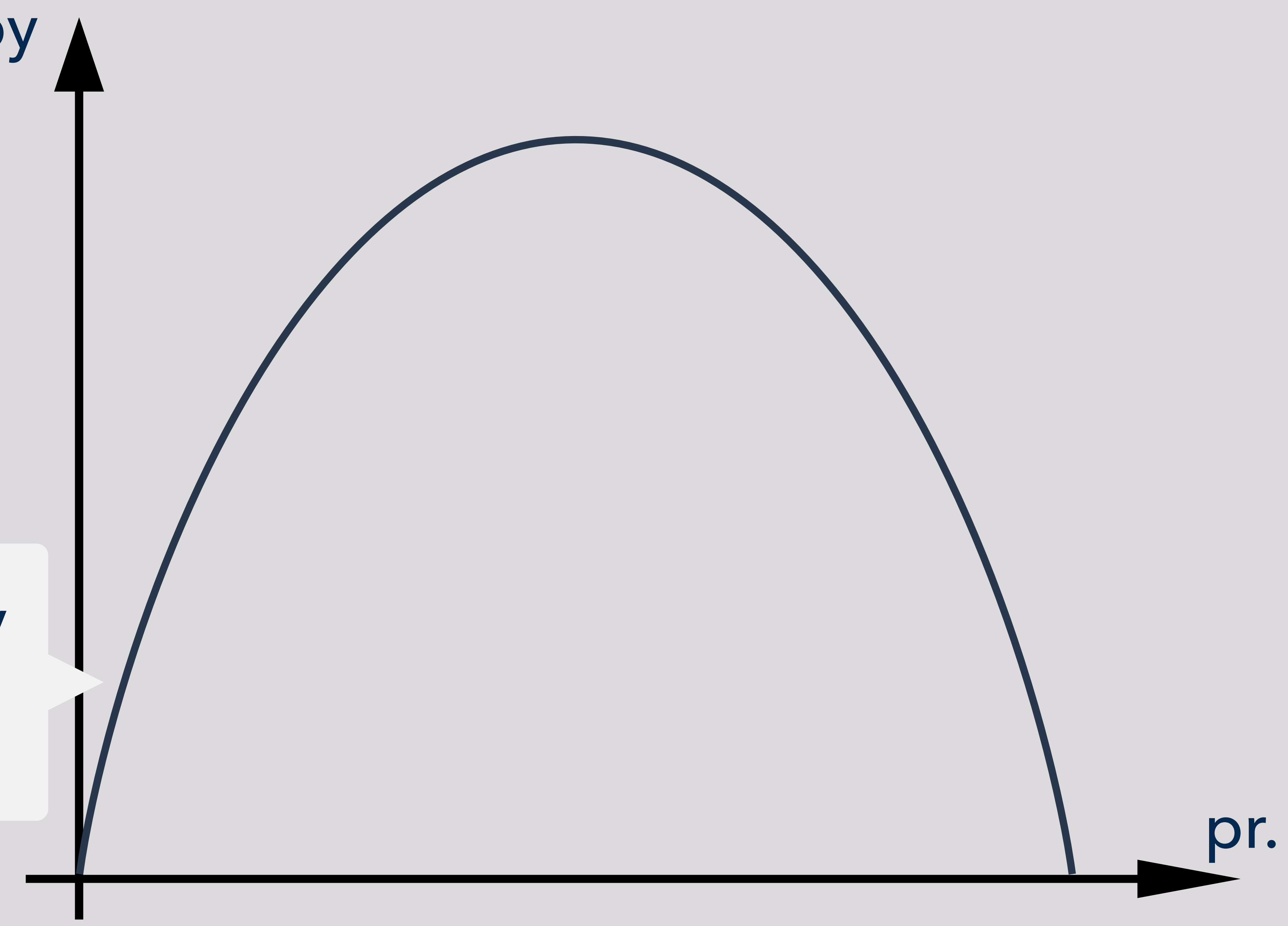
θ

info

Agent i optimizes over cor

entropy measures uncertainty

$\mathbb{H}(X) = -\sum_x \Pr(x)\log(\Pr(x))$



Information cost

entropy cost $=$ unit cost λ $\times$ uncertainty reduction in terms of entropy

$= 1$

$=$ prior entropy $-$ $\mathbb{E}$[posterior entropy]

more

state

θ

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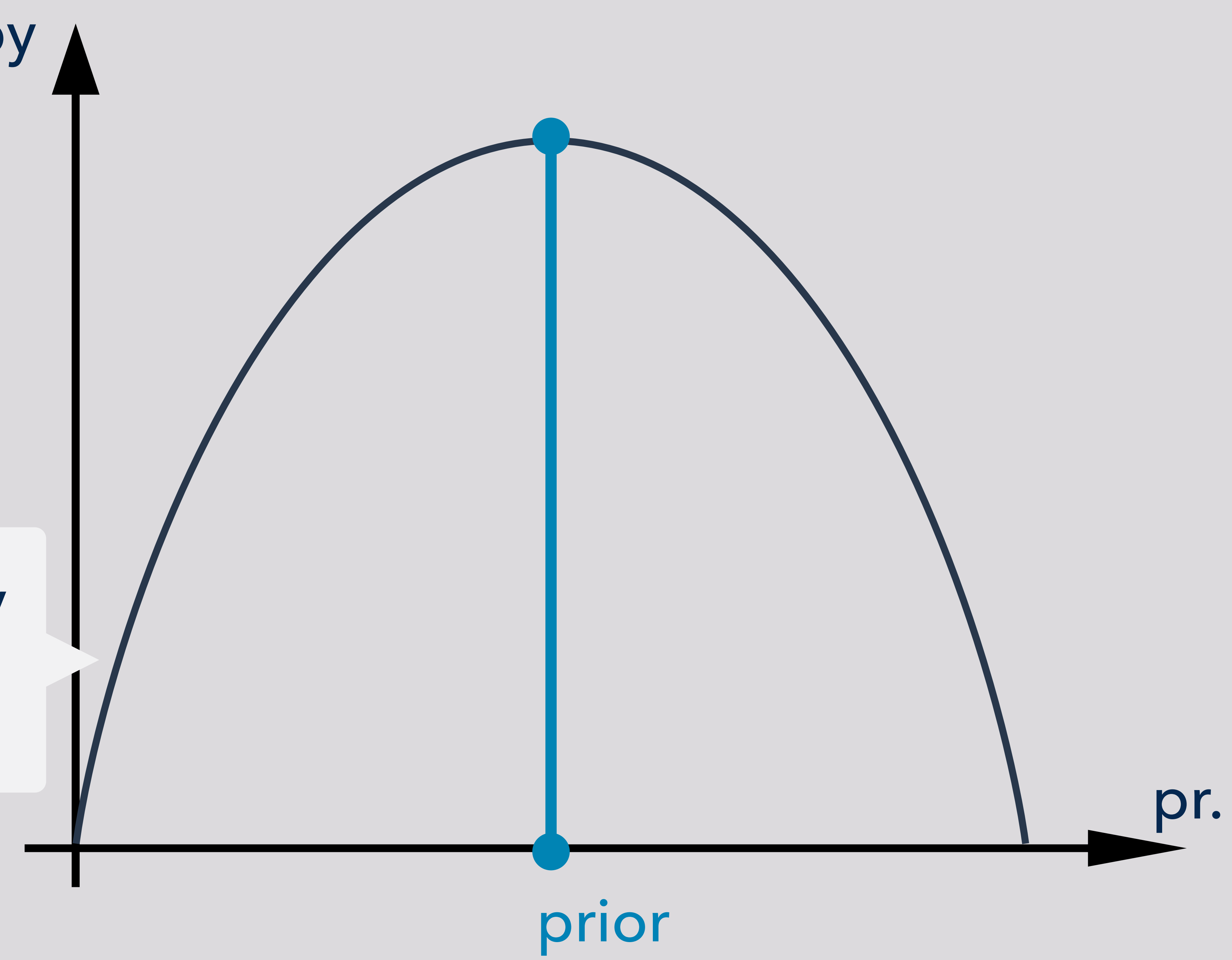

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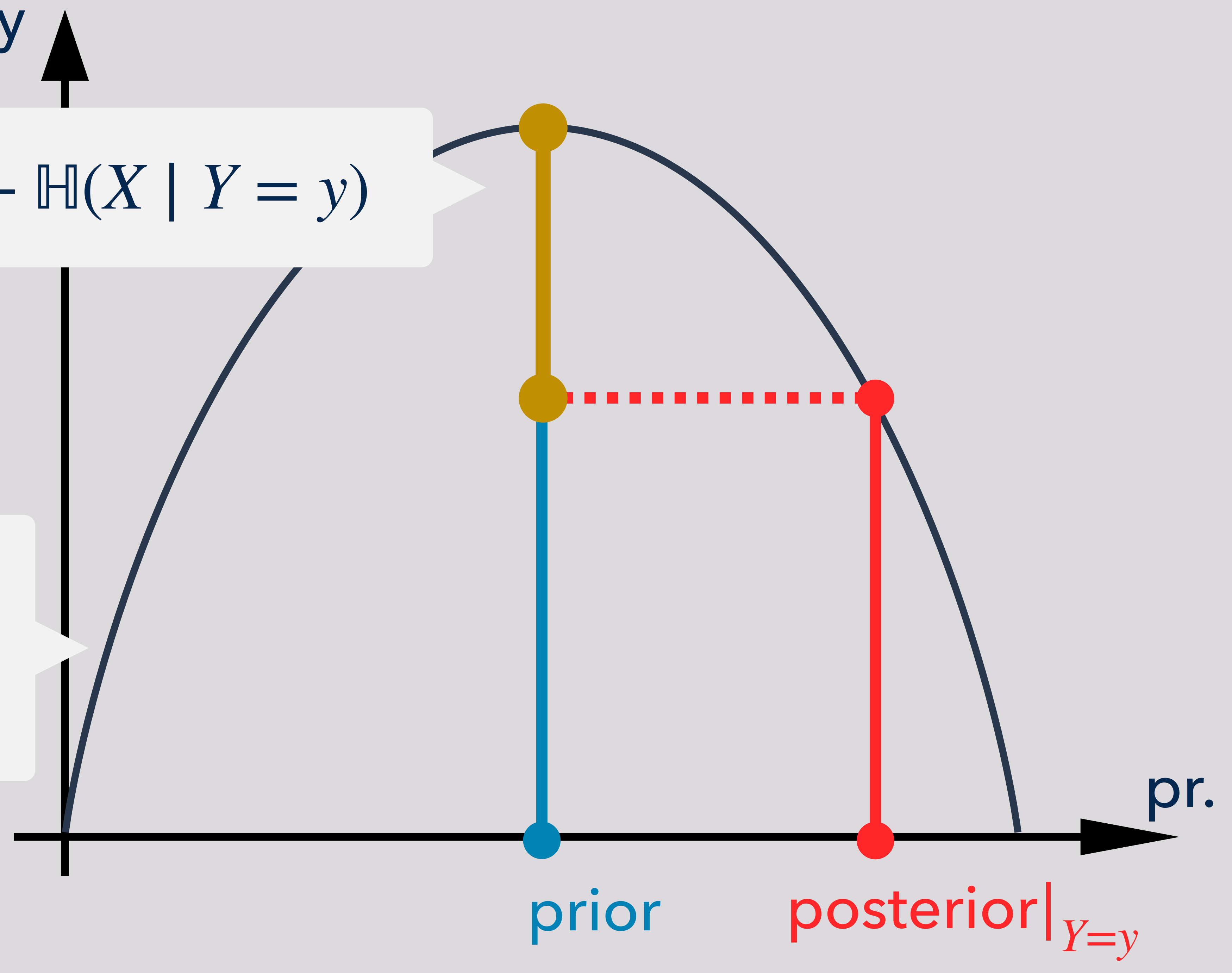

entropy reduction $\mathbb{H}(X) - \mathbb{H}(X \mid Y = y)$

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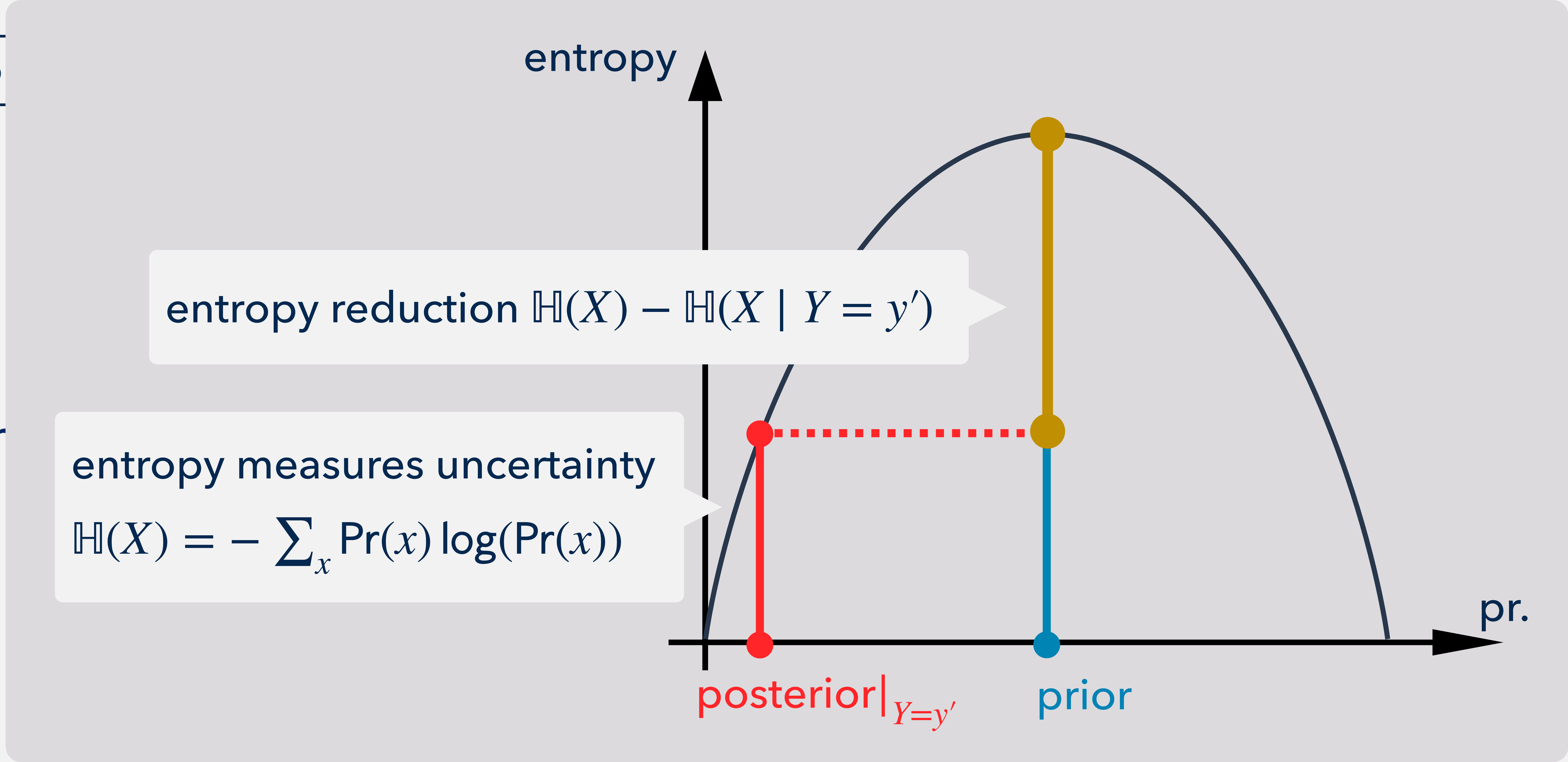

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more

state θ $\xrightarrow{\text{information}}$ **actions** $a_1, \dots, a_n$ $\xleftarrow{\text{information}}$ **average action** $\bar{a}_{-i} = \frac{1}{n-1} \sum_{j \neq i} a_j$

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more

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θ

information →

actions

$a_1, \ldots, a_n$

← information

aggregate →

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more

strategy

$P_i(a_i \mid \bar{a}_{-i}, \theta)$

belief

$\pi_i(\bar{a}_{-i}, \theta)$

avg action & state

$(\bar{a}_{-i}, \theta)$

Equilibrium Denti 2023 Hébert–La'O 2023

strategy

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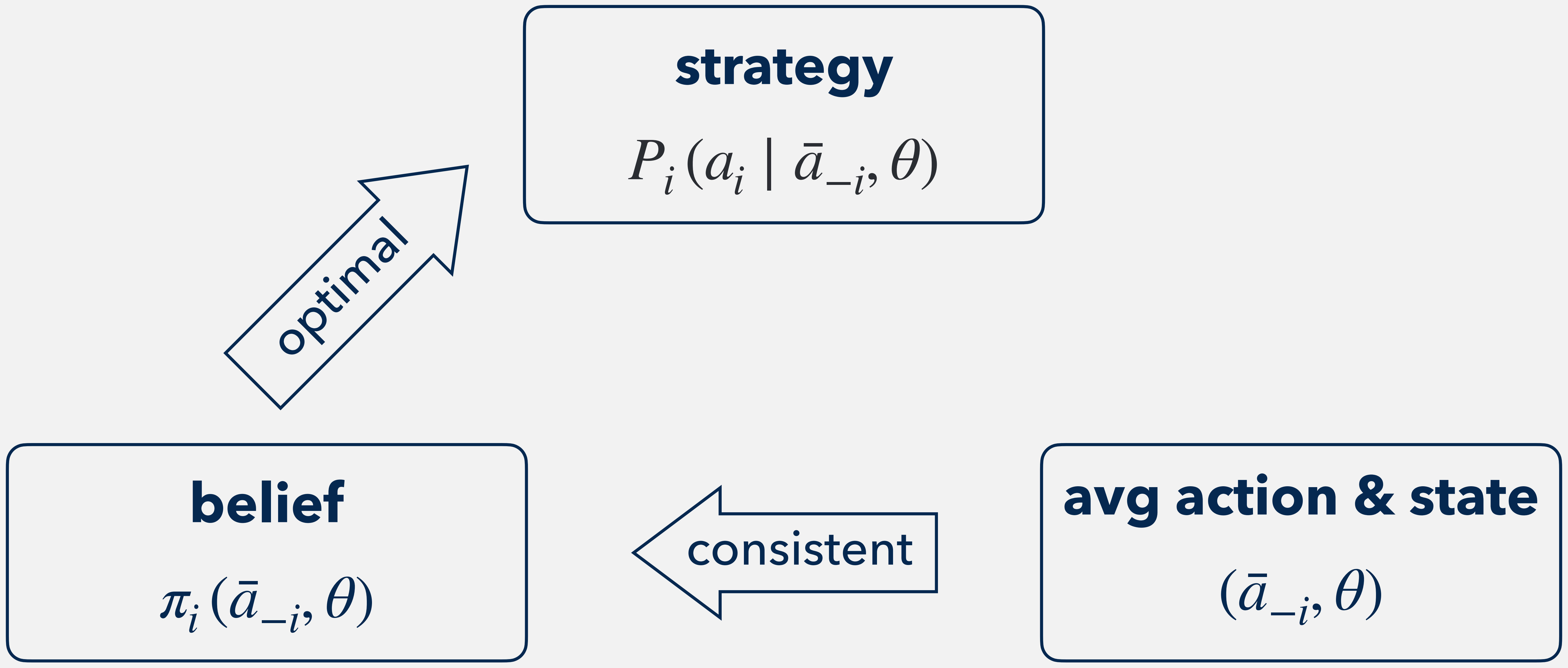
strategy
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optimal
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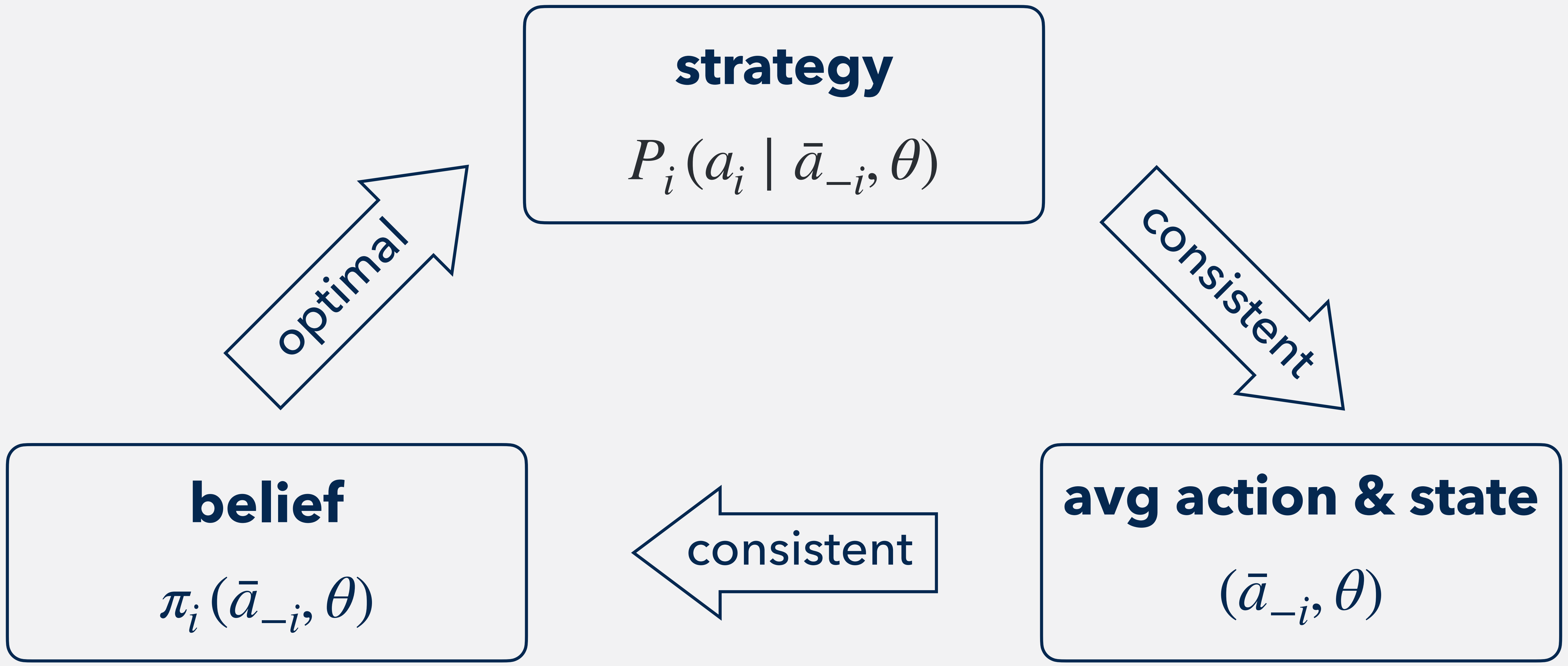
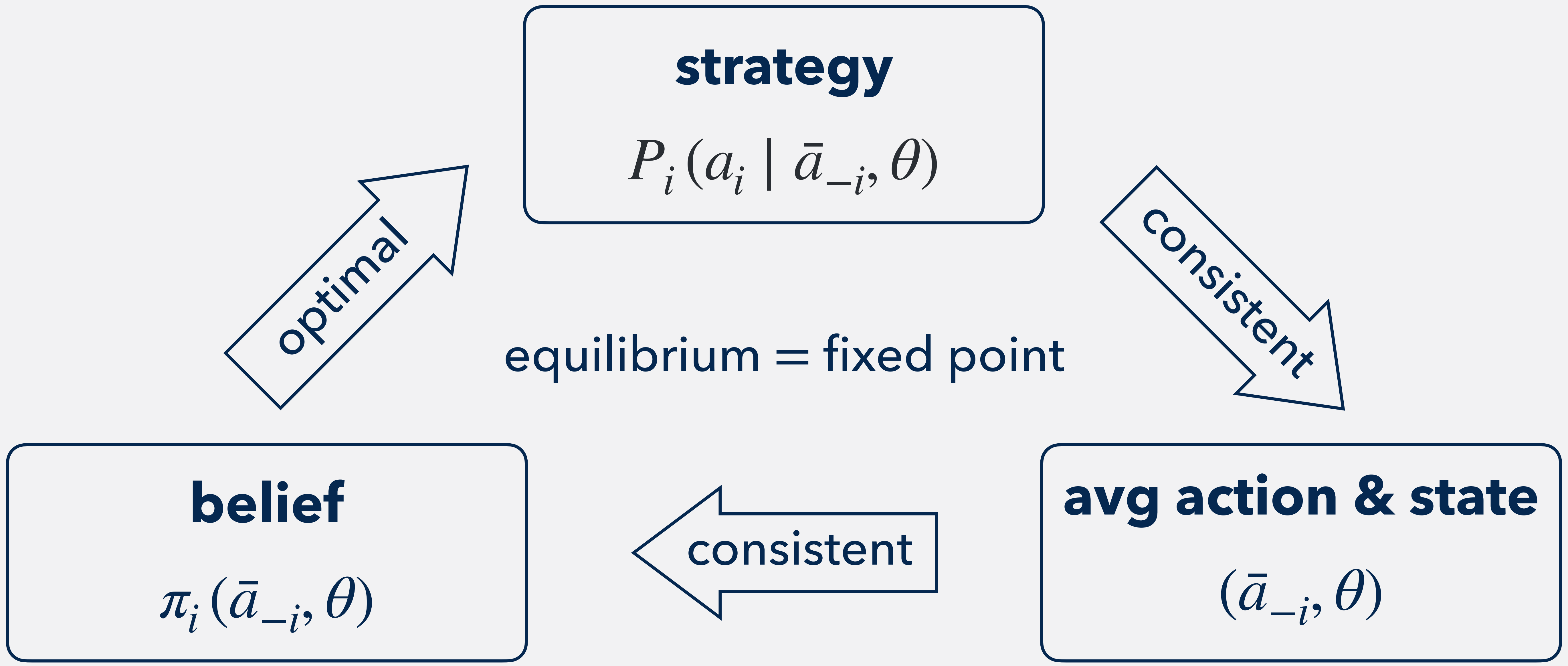
Equilibrium Denti 2023 Hébert–La'O 2023

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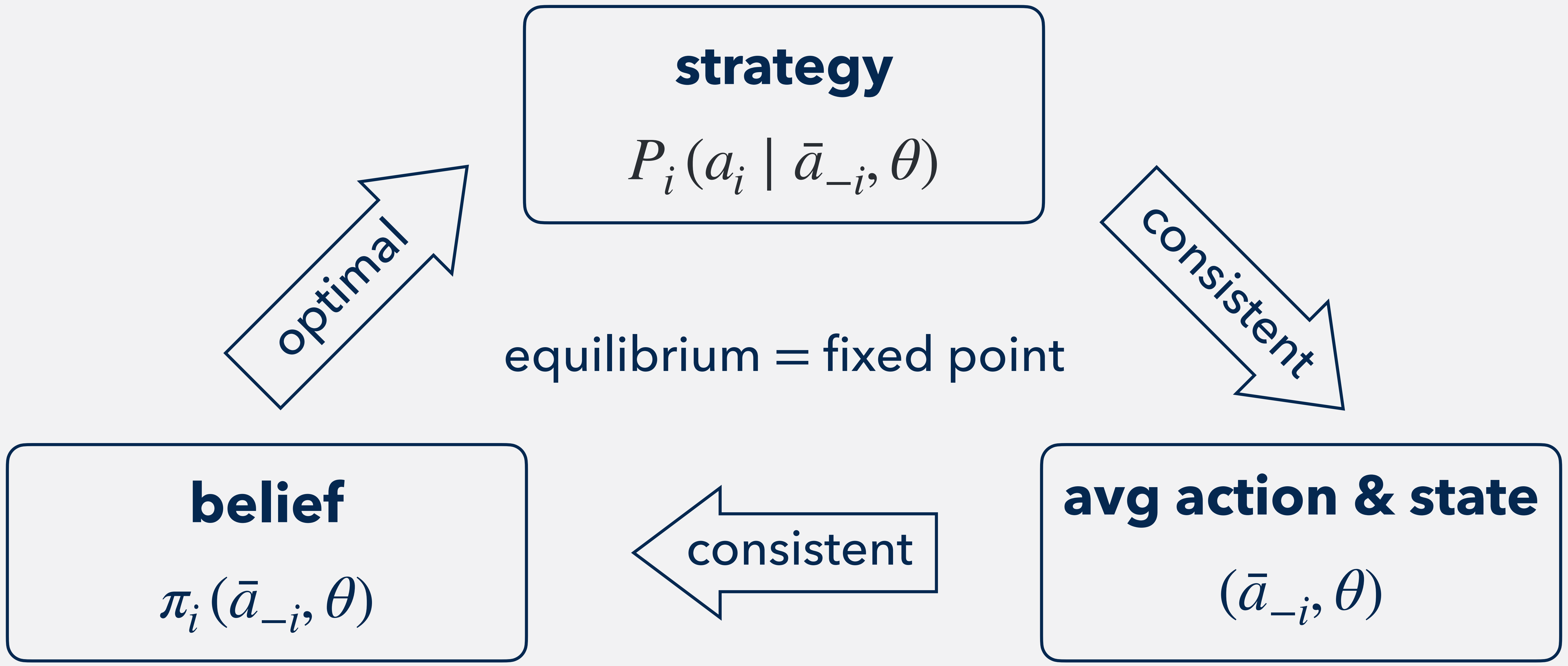
optimal
equilibrium = fixed point



An **equilibrium** is a joint distribution $P_n^*(a_1, \ldots, a_n, \theta)$ such that:

- conditional distribution $P_n^*(a_i \mid \bar{a}_{-i}, \theta)$ = agent i's best response
- marginal distribution of state θ = prior π

focus on symmetric equilibria

A blend of two equilibrium concepts

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- **Bayesian Nash Equilibrium**
  - agents optimize under uncertainty

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  - agents condition their behavior upon an endogenous average action

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Grossman–Stiglitz (1980) etc.

- agents condition their behavior upon an endogenous average action (e.g., prices)

Equilibrium = Biased Logit Matějka–McKay 2015 Caplin–Dean–Leahy 2021 Denti 2023

$$P_n^*(a \mid \theta) \quad \propto \quad \exp\left[v_n(a,\theta)\right] \left(y_n^*\right)^{n\bar{a}_n}\left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$$

more

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potential v_n

more

y_n^* is the marginal probability of action 1

$$y_n^* = \sum_{a_{-i},\theta} P_n^*(a_i = 1, a_{-i}, \theta)$$

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potential v_n

$n\bar{a}_n =$ #agents choosing action 1

- joint $P_n^*(a \mid \theta)$ and marginal y_n^* are interdependent
  - exact logit $P_n^*(a \mid \theta) \propto \exp[v_n(a, \theta)] \iff y_n^* = 1/2$

more

Aggregation

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \quad \equiv \quad \sum_{a:\bar{a}_n \approx x} P_n^*(a \mid \theta)$$

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$\exp\left[v_n(a,\theta)\right]\left(y_n^*\right)^{n\bar{a}_n}\left(1-y_n^*\right)^{n(1-\bar{a}_n)}$

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$$\approx \exp\left[v_n(x,\theta)\right] \sum_{a:\bar{a}_n \approx x} \left(y_n^*\right)^{n\bar{a}_n}\left(1-y_n^*\right)^{n(1-\bar{a}_n)}$$

Aggregation

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \equiv \sum_{a:\bar{a}_n \approx x} P_n^*(a \mid \theta)$$

$P_n^*(a \mid \theta) \propto$
$\exp\left[v_n(a,\theta)\right] \left(y_n^*\right)^{n\bar{a}_n}\left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$

$$\propto \sum_{a:\bar{a}_n \approx x} \exp\left[v_n(a,\theta)\right] \left(y_n^*\right)^{n\bar{a}_n}\left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$$

$v_n(a,\theta)$ depends on average $\bar{a}_n \approx x$

$$v_n(a,\theta) = n\left[\frac{\bar{a}_n^2}{2} - \theta\bar{a}_n\right] + \frac{\bar{a}_n}{2}$$

$v_n(x,\theta)$ is continuous in x

$$\approx \exp\left[v_n(x,\theta)\right] \sum_{a:\bar{a}_n \approx x} \left(y_n^*\right)^{n\bar{a}_n}\left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$$

potential factor **likelihood factor**

Potential Factor

$$\exp\left[\, v_n(x,\theta) \,\right] = \exp\left[n \cdot \frac{v_n(x,\theta)}{n} \right] \approx \exp\left[n\, f(x,\theta) \right]$$

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$$v_n(x,\theta) = n\left[\frac{x^2}{2} - \theta x\right] + \frac{x}{2}$$

The **limit potential**

$$\frac{v_n(x,\theta)}{n} \xrightarrow{n\to\infty} f(x,\theta) \equiv \frac{x^2}{2} - \theta x$$

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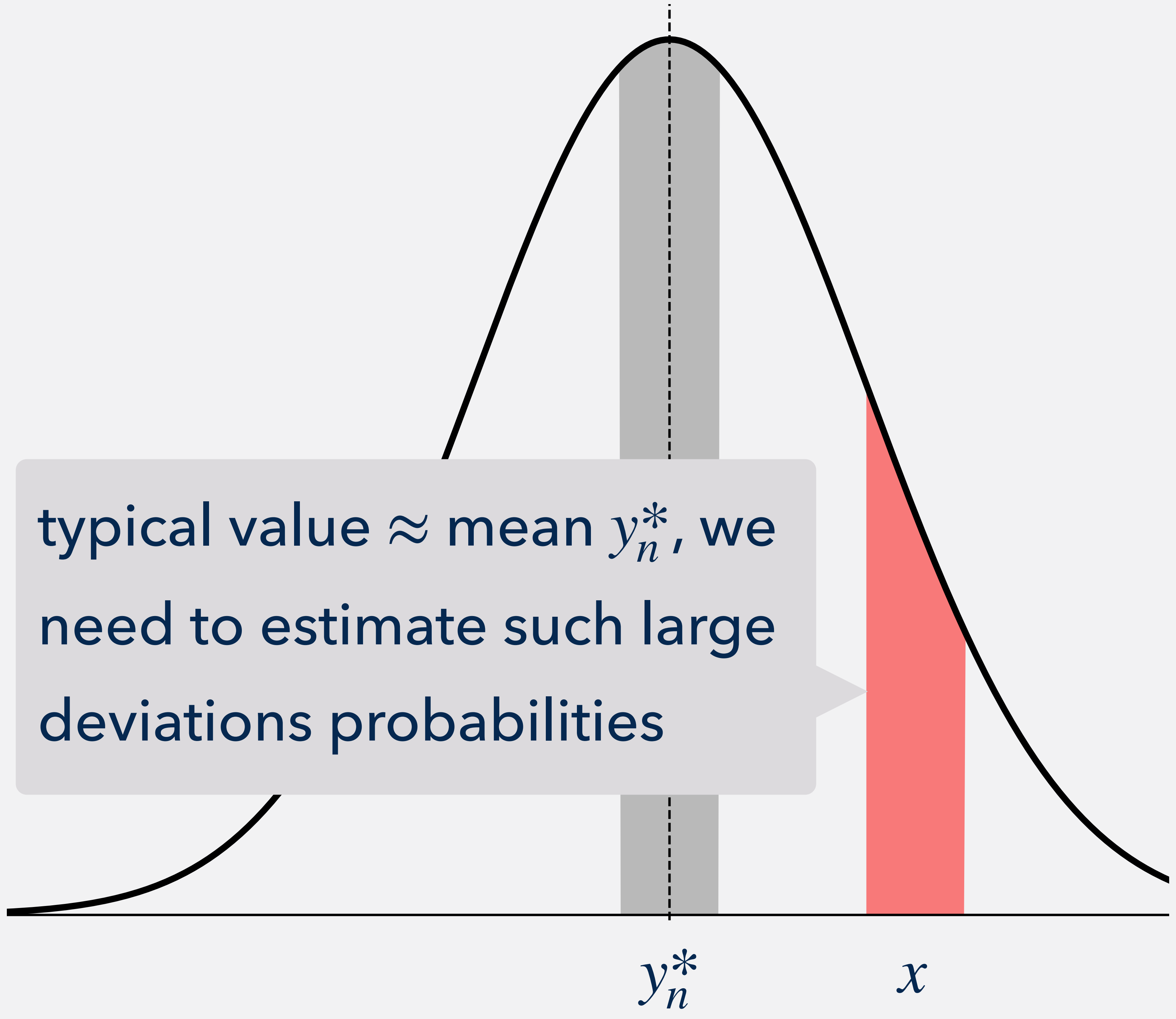
Likelihood Factor

$$\sum_{a:\bar{a}_n \approx x} \left(y_n^*\right)^{n\bar{a}_n}\left(1-y_n^*\right)^{n(1-\bar{a}_n)} = \sum_{k:\frac{k}{n}\approx x} \binom{n}{k} \left(y_n^*\right)^{k}\left(1-y_n^*\right)^{n-k} \approx \exp\left[-n\,\mathrm{KL}\left(x \,\|\, y^*\right)\right]$$



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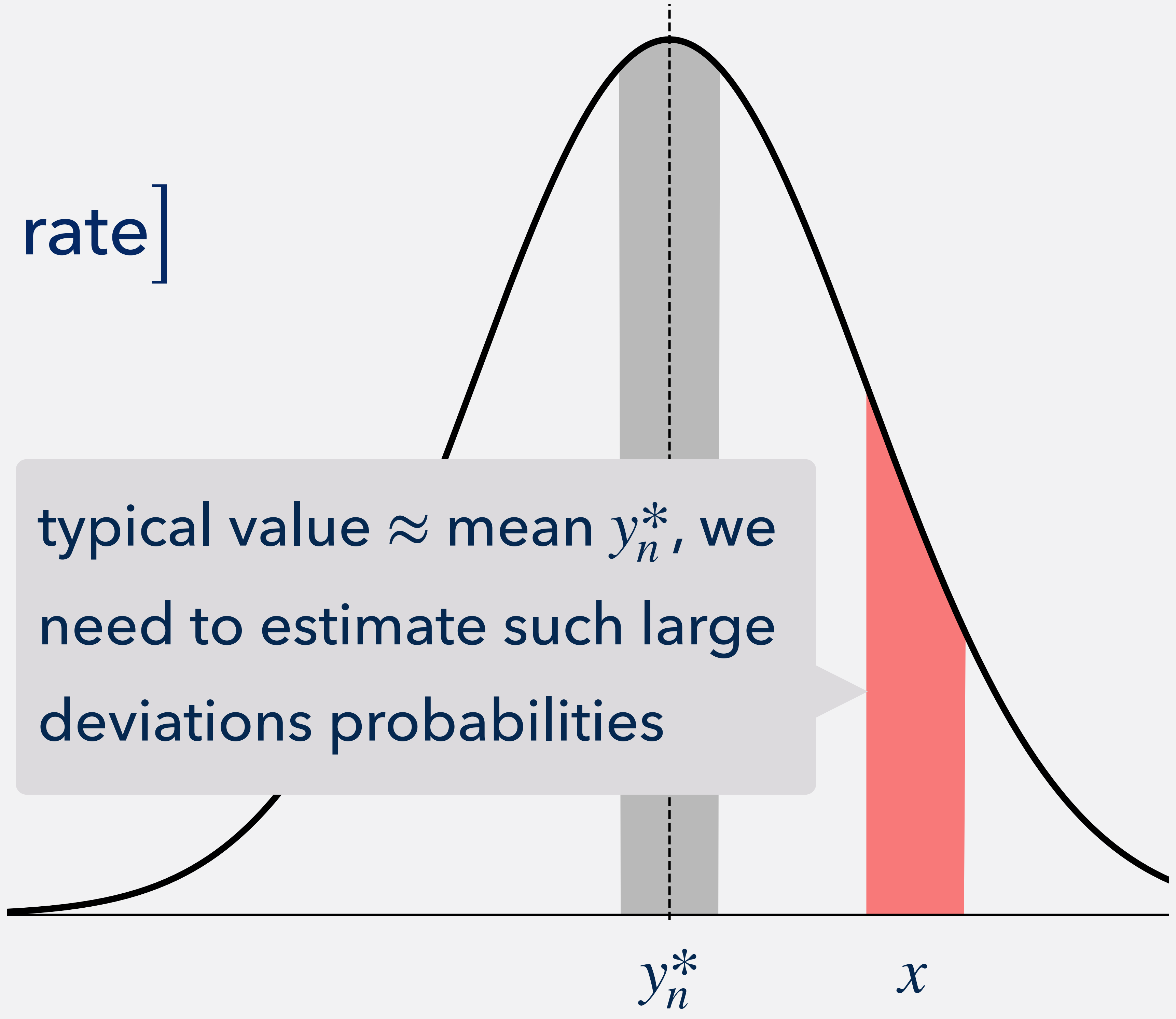
Large Deviations Theory

- how "fast" large deviations probabilities decay:

$$\Pr\left(\text{large deviations}\right) \approx \exp\left[-n \cdot \text{convergence rate}\right]$$



typical value $\approx$ mean y_n^*, we need to estimate such large deviations probabilities

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  - y^* is an accumulation point of $\{y_n^*\}_n$
  - non-iid but KL rate

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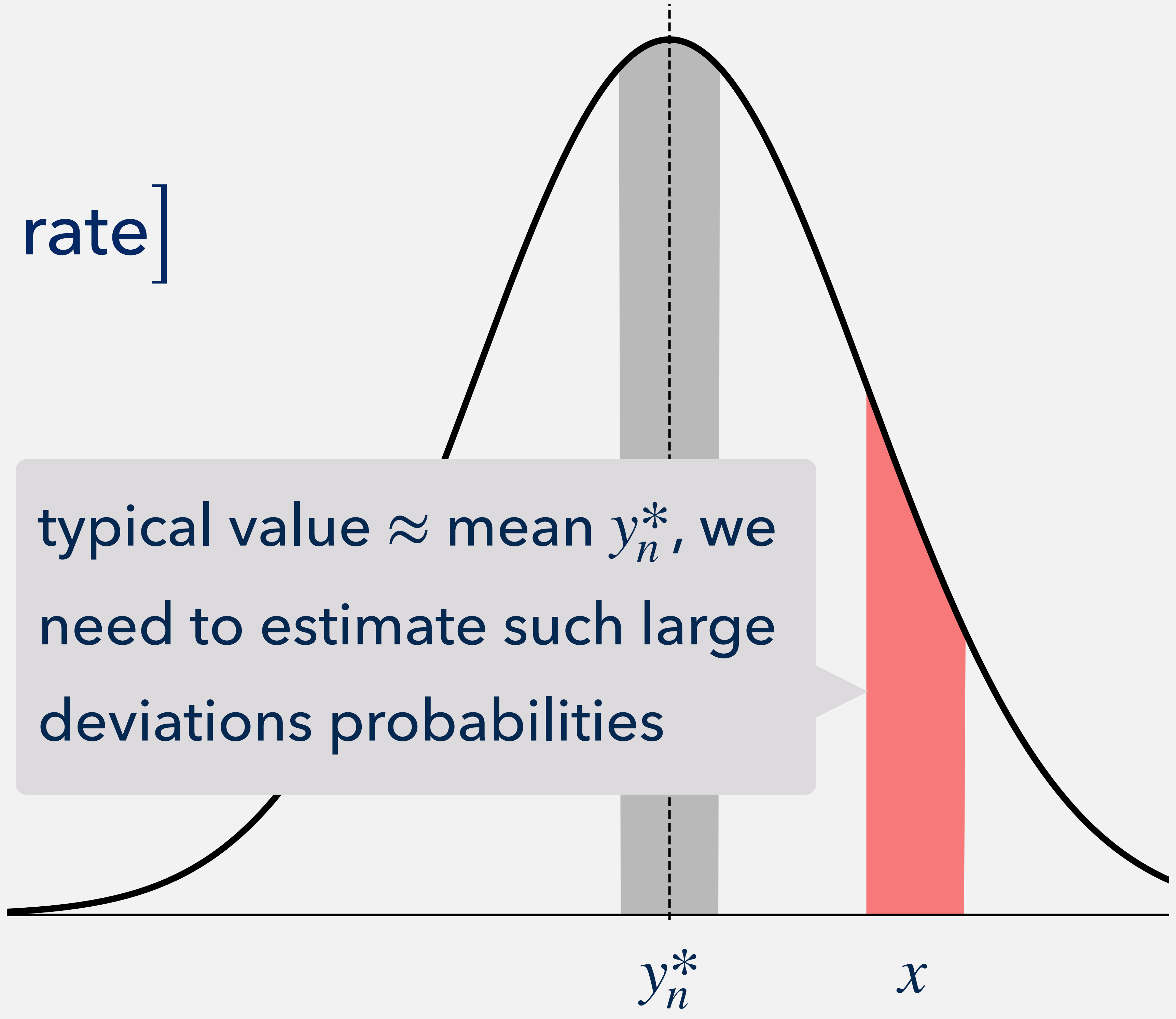

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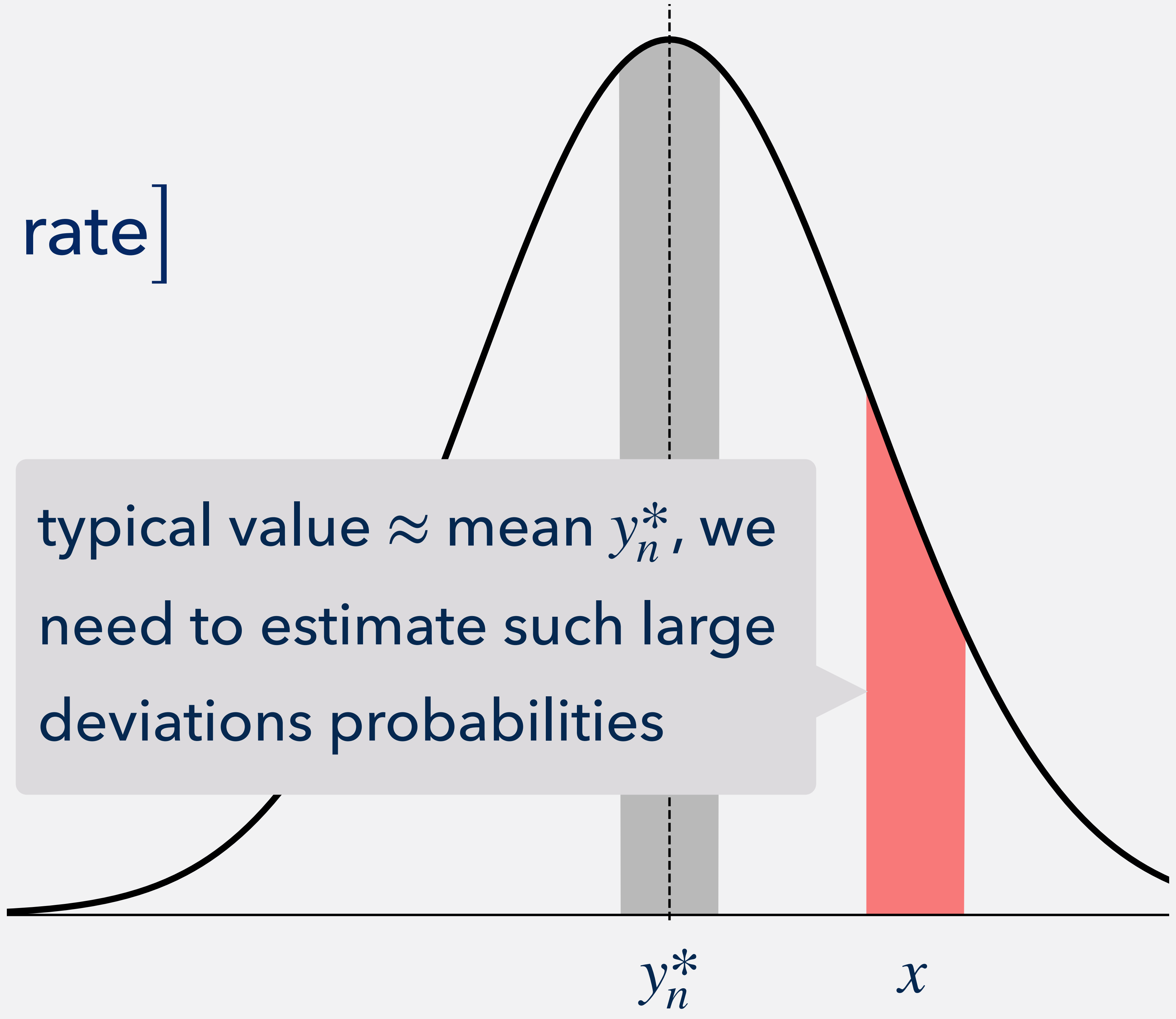

KL-Regularized Potential

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \quad \propto \quad \exp\left[v_n(x,\theta)\right] \sum_{a:\bar{a}_n \approx x} \left(y_n^*\right)^{n\bar{a}_n}\left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$$

potential factor $\approx \exp\left[n\, f\left(x,\theta\right)\right]$

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KL-regularized potential $F\left(x, y^*, \theta\right)$

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \approx \frac{\exp\left[n\, F\left(x, y^*, \theta\right)\right]}{\text{normalizing constant}} \lessapprox \frac{\exp\left[n\, F\left(x, y^*, \theta\right)\right]}{\exp\left[n \max_z F\left(z, y^*, \theta\right)\right]}$$

video

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \approx \frac{\exp\left[n\, F\left(x, y^*, \theta\right)\right]}{\sum_z \exp\left[n\, F\left(z, y^*, \theta\right)\right]} \lessapprox \frac{\exp\left[n\, F\left(x, y^*, \theta\right)\right]}{\exp\left[n \max_z F\left(z, y^*, \theta\right)\right]}$$

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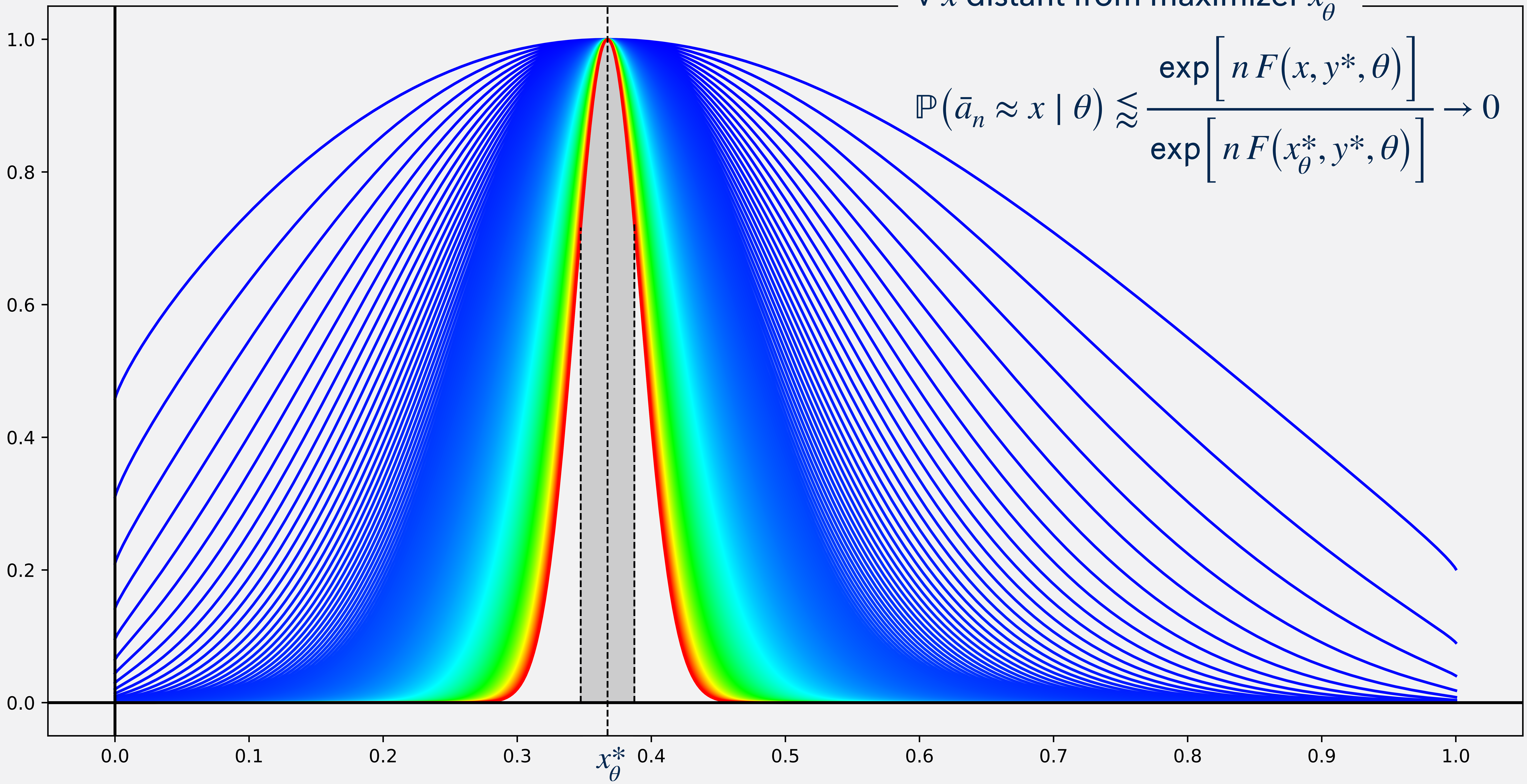
$$x^*_\theta = \arg\max_z F\left(z, y^*, \theta\right)$$

video

$\forall\, x$ distant from maximizer x^*_θ

$$\mathbb{P}\left(\bar{a}_n \approx x \mid \theta\right) \lessapprox \frac{\exp\left[\, n\, F\left(x, y^*, \theta\right)\right]}{\exp\left[\, n\, F\left(x^*_\theta, y^*, \theta\right)\right]} \to 0$$



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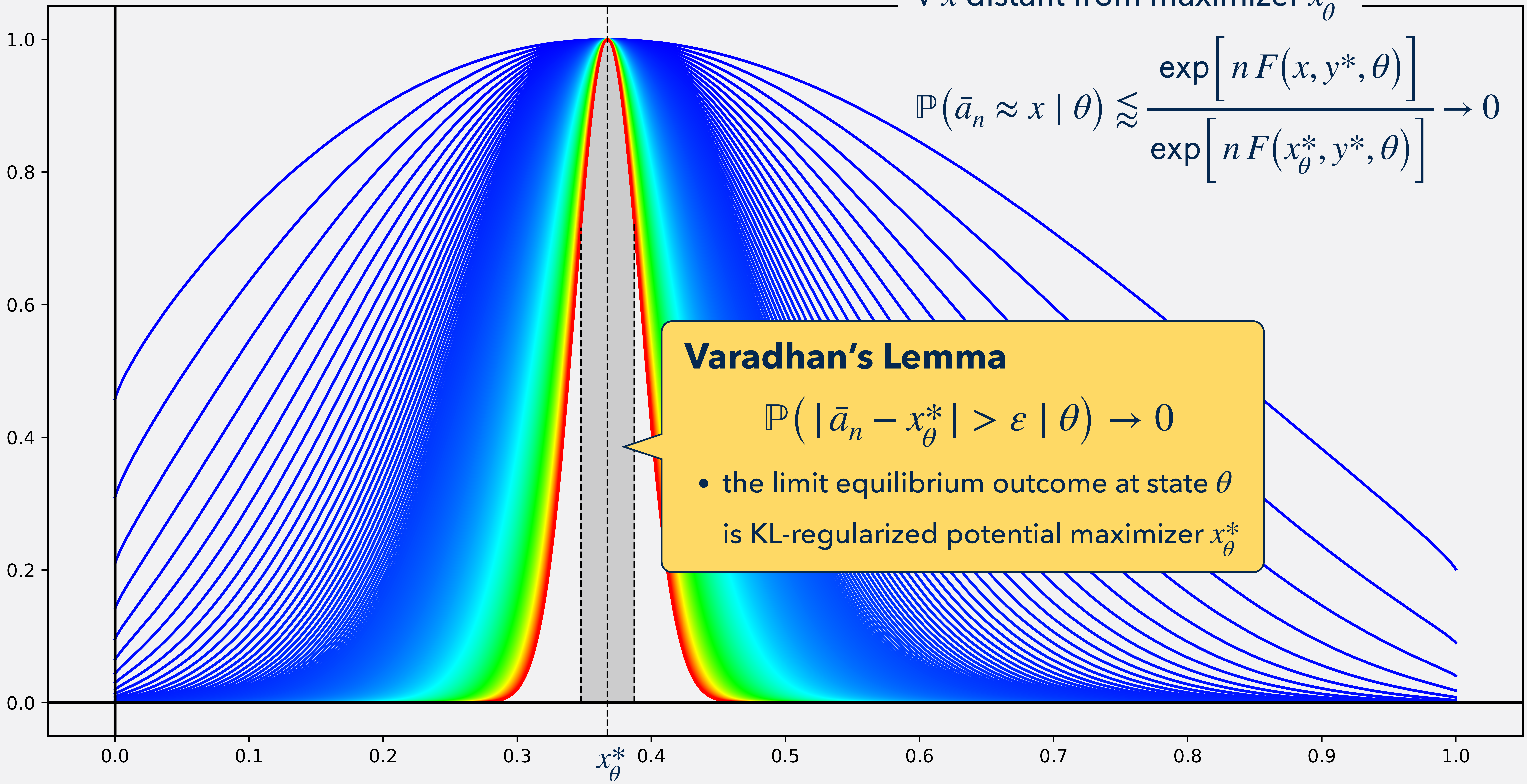

Varadhan's Lemma

$$\mathbb{P}\Big(\,|\bar{a}_n - x^*_\theta\,| > \varepsilon \mid \theta\Big) \to 0$$

- the limit equilibrium outcome at state θ is KL-regularized potential maximizer x^*_θ

more

Main Result 1: Endogenous Volatility

Endogenous volatility **vanishes**

$$\lim_{n\to\infty} \mathsf{Var}\big(\bar{a}_n \mid \theta\big) = \lim_{n\to\infty} \mathsf{Cov}\big(a_i, a_j \mid \theta\big) = 0$$

Aggregate action $\bar{a}_n \xrightarrow{n\to\infty}$ KL-regularized potential maximizer **exponentially fast**

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}\big(\bar{a}_n \in [l, h] \mid \theta\big) = \max_{x \in [l,h]} F\big(x, y^*, \theta\big) - \max_{x \in [0,1]} F\big(x, y^*, \theta\big)$$

This **exact convergence rate** helps estimate

- how many agents are needed to justify the **infinite-agent** approximation

Shortcut to Limit Equilibrium

Limit equilibrium outcome

$$x_\theta^* \in \arg\max_x F\left(x, y^*, \theta\right)$$

hard to compute

- y^* is an accumulation point of $\{y_n^*\}_n$

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Sufficient

$$(x^*, y^*) \in \arg\max_{x,y} \mathbb{E}_\theta\left[f\left(x_\theta, \theta\right) - \mathrm{KL}\left(x_\theta \,\|\, y\right)\right]$$

Main Result 2: Representative Agent

A limit equilibrium outcome satisfies the **representative agent** problem

Necessary: $x^*_\theta \in \arg\max_x F(x, y^*, \theta) \quad \forall\theta, \qquad y^* = \mathbb{E}_\theta[x^*_\theta]$

Sufficient: $(x^*, y^*) \in \arg\max_x \mathbb{E}_\theta[f(x_\theta, \theta)]$ − entropy cost

agent	gross payoff	attention cost
individual	expected	entropy cost (θ, $\bar{a}_{-i}$)
representative	non-expected	entropy cost (θ)

Law of Large Numbers?

We **cannot** use the usual LLN due to correlation in finite-agent models

$$P_n^*(a \mid \theta) \quad \propto \quad \exp\Big[v_n(a,\theta)\Big] \, \big(y_n^*\big)^{n\bar{a}_n}\big(1-y_n^*\big)^{n(1-\bar{a}_n)}$$

- LLN turns out true because $\text{Cov}\big(a_i, a_j \mid \theta\big) \to 0$ by large deviations
- LLN turns out false if KL-regularized potential has multiple maximizers

more

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- LLN turns out false if KL-regularized potential has multiple maximizers

If there is a continuum agents (Hébert–La'O '23)?

- a continuum LLN is "assumed"
- equilibrium set when a continuum agents $\neq$ equilibrium set when #agents $\to \infty$

Example:

- Key modeling features
- Large-deviations logic behind main results

Main Results

- General settings
- Statements

Aggregative Potential Game

State $\theta \sim$ common prior $\pi \in \Delta(\text{finite } \Theta)$

Agent $i = 1, \ldots, n$ chooses action $a_i \in A = \{1, \ldots, K\}$

$u_i(a_i, \text{emp}(a), \theta)$ **potential** $v_n(a, \theta)$ has the **limit potential** $f : \Delta(A) \times \Theta \to \mathbb{R}$

empirical action distribution:

$$\text{emp}(a) = \left(\frac{\#\{i : a_i = k\}}{n} \right)_{k \in A} \in \Delta(A)$$

$$\max_{a \in A^n} \left| \frac{v_n(a, \theta)}{n} - f\left(\text{emp}(a), \theta\right) \right| \xrightarrow{n \to \infty} 0$$

The aggregate action distribution $\mathbb{P}_n^*(\,\cdot \mid \theta) \in \Delta(\Delta(A))$

$$\mathbb{P}_n^*\big(E \mid \theta\big) \equiv \sum_{a:\text{emp}(a)\in E} P_n^*(a)$$

$\equiv \text{Pr}\big(\text{emp}(a) \in E \mid \theta\big)$

Aggregate Action Distribution and KL-Regularized Potential

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every equilibrium is a biased-logit

$$P_n^*(a \mid \theta) \propto \exp\Big[v_n(a,\theta)\Big] \prod_{i=1}^{n} q_n^*(a_i)$$

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$$\approx \exp\left[n\left[f\left(\text{emp}(a),\,\theta\right) - \text{KL}\left(\text{emp}(a) \parallel q^*\right)\right]\right]$$

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KL-regularized potential $F\left(\mathrm{emp}(a), q^*, \theta\right)$

Main Result: Endogenous Volatility Vanishes

Endogenous volatility **vanishes** if $F(\,\cdot\,, q^*, \theta)$ has a unique maximizer

$$\mathbb{P}_n^*\big(O \mid \theta\big) \to 1 \quad \forall \text{open } O \supset \arg\max_p F(p, q^*, \theta)$$

Empirical dist. emp(a) $\xrightarrow{n\to\infty}$ KL-regularized potential maximizer **exponentially fast**

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}_n^*\big(\text{closed } C \mid \theta\big) = \max_{p \in C} F\big(p, q^*, \theta\big) - \max_{p \in \Delta(A)} F\big(p, q^*, \theta\big)$$

This **exact rate of convergence** estimates

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more

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A limit equilibrium outcome satisfies the **representative agent** problem

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agent	gross payoff	attention cost
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representative	non-expected	entropy cost (θ)

Appendix

Suppose that information $\mathtt{i}(\mathbb{P}(E))$ from observing an event E depends on its probability

- $\mathtt{i} : [0,1] \to \mathbb{R}$ is strictly decreasing and continuous
- a probability-1 event has no information: $\mathtt{i}(1) = 0$
- information of two independent E and F equals the sum of the information:

$$\mathtt{i}(\mathbb{P}(E \cap F)) = \mathtt{i}(\mathbb{P}(E)) + \mathtt{i}(\mathbb{P}(F))$$

By these axioms

$$\mathtt{i}(x) = -\log(x)$$

Entropy H is the expectation: $\mathrm{H}(X) = \sum_x \mathbb{P}(x)\mathtt{i}(x)$

back

Marginal Action Distribution

$$P_n^*(a \mid \theta) \quad \propto \quad \exp\left[v_n(a, \theta)\right] \left(y_n^*\right)^{n\bar{a}_n} \left(1 - y_n^*\right)^{n(1-\bar{a}_n)}$$

- y_n^* is the marginal probability of action 1

$$y_n^* = \sum_{a_{-i}, \theta} P_n^*(a_i = 1, a_{-i}, \theta)$$

- y_n^* is a symmetric pure NE of an auxiliary game (Denti '23)
  - each player has an action $y_i \in [0,1]$ and a common payoff function:

$$U(y_1, \ldots, y_n) \equiv \sum_{\theta} \pi(\theta) \log \sum_{a} \exp\left[v_n(a, \theta)\right] \prod_{i:a_i=1} y_i \prod_{i:a_i=0} (1 - y_i)$$

A family $\{\mu_n\}_n$ satisfies the **Large Deviations Principle** if

$$\exists \text{ function } I \text{ such that} \quad \mu_n(E) \approx \exp\left[-n \inf_{x \in E} I(x)\right]$$

- this rate function I gives the rate of convergence
  - example: $I(x) = \mathrm{KL}(x \,\|\, y^*)$

Varadhan's Lemma:

$$\int \exp\Big[n\, g(x)\Big] \mu_n(\mathrm{d}x) \approx \exp\Big[n \sup_x \{g(x) - I(x)\}\Big] \quad \forall \text{continuous, bounded } g$$

- Sanov's theorem is a special case of $g \equiv 0$

back

Varadhan's Lemma

A family $\{\mu_n\}_n$ satisfies the **Large Deviations Principle** if ∃ lower semicontinuous I such that

- for each k, the set $\{x : I(x) \le k\}$ is compact
- for each Borel B

$$-\inf_{x \in \mathrm{int}(B)} I(x) \le \liminf_{k\to\infty} \frac{1}{n} \ln \mu_n(B) \le \limsup_{k\to\infty} \frac{1}{n} \ln \mu_n(B) \le -\inf_{x\in \mathrm{cl}(B)} I(x)$$

Varadhan's lemma: For any continuous and bounded g,

$$\limsup_{n\to\infty} \frac{1}{n} \log \mathbb{E}_{\mu_n}\left[\exp\left[n\, g(X)\right] \mathbf{1}_C(X)\right] \le \sup_{x\in C}\{g(x) - I(x)\} \qquad \forall \text{closed } C$$

$$\liminf_{n\to\infty} \frac{1}{n} \log \mathbb{E}_{\mu_n}\left[\exp\left[n\, g(X)\right] \mathbf{1}_O(X)\right] \ge \sup_{x\in O}\{g(x) - I(x)\} \qquad \forall \text{open } O$$

- Sanov's theorem is a special case of $g \equiv 0$

back

Correlated Information vs Independent Information

As #agents $n \to \infty$, the equilibrium is such that

$$\textbf{correlated information} \to \textbf{independent information}$$

- **correlated** information
  - an exogenous state θ and an endogenous average $\bar{a}_n$
- **independent** information
  - an exogenous state θ

back

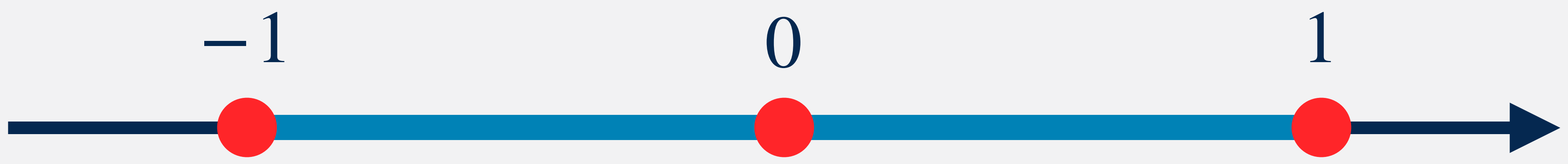
Continuum-Agent Equilibria $\subsetneq$ $(n \to \infty)$-Agent Equilibria

Majority-voting game

- state $\theta = \pm 1$ with equal prior $1/2$
- agent i chooses action $a_i = \pm 1$

$$\text{payoff } u_i(a, \theta) = \begin{cases} 1 & \text{if } \operatorname{sgn}(\bar{a}_n) = \theta \\ 0 & \text{if } \operatorname{sgn}(\bar{a}_n) \neq \theta \end{cases}$$

- equilibrium



"anything goes" in the continuum-agent case

- equilibrium $\bar{a} \in [-1,1]$
- all **red**/**blue** outcomes are possible

"something goes" in the $(n \to \infty)$-agent case

- equilibrium $\lim_n \bar{a}_n \in \{\pm 1, 0\}$
- only **red** outcomes are possible

back

> Information-processing noise that is independent across agents will average out in macroeconomic behavior, whereas **highly correlated information processing noise will become an additional source of macroeconomic randomness**
>
> Sims 2010

Correlated noise...

- **vanishes** when #agents $n \to \infty$ if KL-regularized potential has a unique maximizer
- **persists** when #agents is large but finite
  - the rate of convergence helps estimate this effect

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